MIL-S-21516B (Wep)
28 May 1963
Superseding
MIL-S-21516A (Wep)
4 April 1961

MILITARY SPECIFICATION

SUPPORT RING ASSEMBLY LAUNCHER TUBE, MISSILE

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope - This specification states the requirements for the support ring assembly for Launcher Tube, Missile, Mark 4 Mods 0, 1, 2, and 3 and Mark 8 Mods 0 and 1,

1.2 Classification - The support ring assembly shall be classified as follows:

Type I - Pneumatic operation

Type II - Hydraulic operation

2. APPLICABLE DOCUMENTS

2.1 Documents - The following documents of the issue in effect on date of invitations for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-05272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

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2.1 (Cont)

NAVY DEPARTMENT

BUREAU OF SHIPS	General Specifications for Ships of the United States Navy, Section S9-1, Welding and Allied Processes

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS, Code Ident 10001

LD 419544	Support Ring Assembly, Missile (for Type I)
LD 540975	Support Ring Assembly, Missile (for Type II)
Dwg 2025362	Pneumatic High Pressure Test Fixture
Dwg 2025374	Pressure Tank
Dwg 2244646	Supports
SA-2440143	Simulated Missile Skirt
SA-2440205	Structural Test Setup
SA-2440379	Corrosion Tank

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2.1 (Cont)

PUBLICATIONS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS

NAVWEPS OP 400	General Instrutions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment
NAVWEPS OP 2230	Workmanship and Design Practices for Electronic Equipment
NAVORDINST 4355.1	Equipment fOr use by Government Inspectors in Final Inspection of Naval Ordnance Items
BUWEPSINST 4355.5	Bureau of Naval Weapons Calibration Program - Standards Laboratory Information Manual

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other Publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 — Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committees 202 Chicago Union Station, Chicago 6, Illinois.)

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3. REQUIREMENTS

3.1 Description - The support ring assembly forms a restraining support for the missile during stowage in the launcher tube. It consists of a support ring, clamping ring, clamping ring actuator electrical switches to indicate clamping ring position and switches to indicate first motion of the missile.

3.1.1 Components - The support ring assembly shall COnSist of the components shown on BuWeps LD 419544 for Type I and BuWeps LD 540975 for Type II.

3.2 General Requirements

3.2.l Preproduction Samples - Unless otherwise specified in contract or order, preproduction samples of the support ring assembly shall be manufactured using the methods and procedures proposed for the production lot. These sampies will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification (refer to paragraph 4.2.1). At the discretion of the procuring agency, the requirement for a preproduction sample may be waived and shall then be satisfied by substitution of a production unit chosen at random by the Government Inspector.

3.2.2 Materials - Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the support ring shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BuWeps LD 419544 or 540975.

3.2.3 Finishes, Coatings, Construction, and Dimensions - Finishes, coatings, construction, and dimensions for the support ring assembly shall be in accordance with the apppli-cable drawings, specifications, and publications listed in BuWeps-LD 41954 or 540975.

3.2.4 Marking - Marking of assemblies, subassemblies, and parts shall be in accordance with applicable drawings listed in BuWeps LD 429544 or 540975.

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3.2.5 Interchangeability - Individual assemblies and piece parts of assemblies fabricated in agreement with their applicable drawings and specifications shall be interchangeable with like numbered parts.

3.2.6 Gages - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, requisition, or order, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final inspection Gages.

3.2.7 Threads - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Request for Deviations - The procedures for requesting deviations from this specification or any specification, publication, material, or process specified herein shall be as set forth in NAVWEPS OP 400.

3.2.9 Conflicting Requirements - In event of conflict between this specification and any specification, publication, or drawing referenced herein, this specification shall govern.

3.2.10 Tolerances - Unless otherwise specifically noted, tolerances and accuracies stated herein are absolute.

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3.3 Performance Requirements

3.3.1 Operation

3.3.3.1 Clamping Ring - The clamping ring, when operated by the clamping ring actuator, shall engage the missile skirt such that at least 50 percent of the circumference in each quadrant of the ring contacts the skirt.

3.3.1.2 Clamping Ring Actuator

3.3.1.2.1 Type I - The clamping ring actuator shall have double-acting cylinders such that upon application of 3,800 to 4,500 psig pneumatic pressure to the clamping side or the cylinder the ring will close. Application of the pressure to the unclamping side of the cylinder will open the ring. In case of pressure failure, the clamping ring will open by spring action.

3.3.1.2.2 Type II - The clamping ring actuator shall have double-acting cylinders such that Upon.application of 3,000 $\pm$ 150 psig hydraulic pressure to the clamping side of the cylinder the ring will Close. Application of the pressure to the unclamping side of the cylinder will open the ring. In case of pressure failure the clamping ring will open by spring action.

3.3.1.3 Clamping Ring Position Indicator - The clamping ring position indicator shall provide electrical signal to indicate when the missile skirt is engaged or when it is disengaged.

3.3.1.4 Departure Switches - The departure switches shall initiate an electrical signal when the switch rod is at a point between 0.040 and 0.125 inch above the missile support surface.

3.3.2. Dielectric Strength - The electrical components on the supporting assembly shall withstand a test voltage of 325-volts AC (rms), for 1 minute, applied between common terminals and a ground, and shall show no damage.

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3.3.3 Insulation Resistance - The electrical components on the support ring assembly shall have an insulation resistance of not less than 3 megohms when a potential of 500-volts DC is applied between a terminal and ground with the part at normal operating temperature.

3.3.4 Continuity - The electrical components shall provide continuous circuits between terminal points.

3.3.5 Working Pressure.

3.3.5.1 Type I - The clamping ring actuator and pneumatic piping shall withstand a maximum internal working pressure of 4,550 psig and shall not leak at a rate greater than 2 standard cubic inches per hour (1 cubic inch per cylinder).

3.3.5.2 Type II - The clamping ring actuator and piping shall withstand a maximum internal working pressure of 3,000 ± 150 psig for 1 hour and shall show no leakage. Wetting of the surface shall not constitute leakage. Leakage not exceeding 5 cubic centimeters per cylinder is permissible during 1,000 cycles of operation.

3.6 Hydrostatic Pressure

3.3.6.1 External - The support ring assembly shall withstand an external hydrostatic pressure of not less than 580 psig without leakage or damage.

3.3.6.2 Internal

3.3.6.2.1 Type I - The pneumatic components of the support ring assembly, except the hoses, shall withstand an internal hydrostatic test pressure of 6,750 ± 50 psig for 2 minutes; the hoses shall withstand an internal hydrostatic. test pressure of 7,500 ± 50 psig for 2 minutes. The components shall show no leakage or damage.

3.3.6.2.2 Type II - The hydraulic components of the support ring assembly including the hoses shall withstand an internal hydrostatic test pressure of 4,500 ± 100 psig for 10 minutes and shall show no leakage or damage.

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3.3.7 Operating Life

3.3.7.1 Clamping Ring, Type I - The clamping ring, when operated as specified in 3.3.1.2.1, shall have an operating life of 1,000-cycles from disengaged to engaged to disengaged position without maintenance, and thereafter shall show no signs of wear detrimental to continued operation.

3.3.7.2 Clamping Ring, Type II - The clamping ring, when operated as specified in 3.3.1.2.2, shall have an operating life of 2,000 cycles from disengaged to engaged to disengaged position without maintenance, and thereafter shall show no signs of wear detrimental to continued operation. Leakage not exceeding 10 cubic cetimeters per cylinder is permissible during 2,000 cycles of operation.

3.4 Environmental Conditions

3.4.1 Vibration, Operating Condition - The support ring assembly shall withstand steady-state vibration defined by Type I of Standard MIL-STD-167 in the range of 5 to 25 cycles per second, shall function during and after such vibrations, shall show no damage and thereafter shall be capable of meeting all performance requirements of this specification.

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3.4.2 Vibration, Shipping Condition - The support ring assembly, packaged and packed in accordance with section 5 herein, shall withstand the following vibration requirements and thereafter shall be capable of meeting all performance requirements of this specificaiton:

VIBRATION CRITERIA FOR PACKAGING (MAX)

Type of Transportation	Vertical cps	Vertical g	Longitudinal cps	Longitudinal g	Lateral cps	Lateral g
Truck	200 10	2.5 1.0	150 10	1.5 1.0	200 10	1.0 1.0
Rail	60 10	2.0 1.0	60 10	1.0 0.5	30 10	1.0 0.5
Aircraft	60-100 100-200	1.0 0.5	60-100	1.0	60 100	1.0 0.5
Ship (surface)	10 30	0.5 1.5	Negligible		10 30	0.25 0.75

3.4.3 Shock, Operating Condition - With a missile installed, the support ring assembly shall withstand loading caused by missile acceleration of 5 g downward and 3 g upward (including gravity) axially and 4 g transversely, shall show no damage and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.4 Inclination, Support Ring Assembly - The support ring assembly shall function satisfactorily by holding the missile clamped while withstanding inclination corresponding to a ship's roll of 60 degrees to each side of the vertical during a period of 15 seconds, for a complete cycle of 120 degrees, shall show no malfunction and thereafter shall be capable-of meeting all performance requirements of this specification.

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3.4.5 Corrosive Atmosphere - The support ring assembly shall withstand the salt spray test specified in Procedure I of Specification MIL-E-5272, thereafter shall show no corrosion detrimental to operation, and shall be capable of meeting all performance requirements of this specification.

3.4.6 Submersion - The support ring assembly shall withstand submersion in sea water for 90 consecutive days at temperatures between +28 and +100 degrees Fahrenheit, shall show no damage, and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.7 Heat/Pressure Cycle

3.4.7.1 Support Ring Assembly, Type I - Not applicable

3.4.7.2 Support Ring Assembly, Type II - The Type II support ring assembly shall withstand the environment of 5 consecutive steam heat/pressure cycles without damage, and thereafter shall be capable of meeting all performance requirements of this specification. A steam heat/pressure cycle shall consist of a change in pressure from atmospheric pressure to 115 ± 15 psig and return to atmospheric pressure coincident with a temperature change from ambient to +325 ± 25 degrees Fahrenheit to +210 ± 20 degrees Fahrenheit within 1 second and then to ambient temperature within 10 minutes.

3.5 Cleanup - Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges shall be removed.

3.6 Workmanship - The equipment, including all parts and accessories shall be constructed and finished in a manner to assure compliance with this specification. Particular attention shall be paid to neatness and thoroughness of plating, painting, welding, wiring, and freedom of parts from burrs and sharp edges. Welding and brazing shall conform to the requirements of section S9-1 General Specifications for Ships of the United States Navy. NAVWEPS QP 2230 Shall be used as a guide for determining acceptable workmanship of electrical parts.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's Inspection - Unless otherwise specified herein, the contractor is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the contractor may tuilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Government Inspection - The Government Inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract, pertinent drawings, and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in NAVWEPS OP 400. Production tests of final assemblies shall be on a 100-percent basis. No sampling is permitted for acceptance inspection purposes. Only those assemblies which pass final inspection shall be accepted by the Navy; a rejected assembly may be reworked and submitted in accordance with NAVWEPS OP 400. Failed parts and, components shall be submitted to the Material Review Board for disposition.

4.2.1 Preproduction Sample - The preproduction sample specified in 3.2.1 shall be submitted to the Government Inspector for preproduction tests at an activity designated in the contract, or by the procuring activity, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order. Disposition of the test sample shall be at the discretion of the Material Review Board.

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4.3 Classification of Tests - The inspection and testing of the support ring assembly shall be classified as follows:

4.3.1 Production Tests - These tests are to be accomplished on each support ring assembly being submitted for acceptance under the contract. Production tests shall be either performed by the manufacturer and witnessed by the Government Inspector, or these tests may be conducted by the Government Inspector. These tests are detailed in 4.5.

4.3.2 Preproduction Tests - Preproduction tests are those which are accomplished on samples selected as specified in 4.2.1 which are representative of the production of the support ring assembly, to determine that the production meets the requirements of this specification. These are detailed in 4.5 and 4.6. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the cessation of production until determination of cause of failure is made and appropriate correction is approved by the procuring activity.

4.3.3 Tests After Delivery - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all of the tests specified herein as deemed necessary after installation of the support ring assembly. Any defects discovered before, during, or after these tests, due to the Contractor's failure to comply with drawings or specifications for this equipment, shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

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4.4 Special Test Equipment - The following items of special test equipment are required to perform the acceptance tests set forth in this specification:

Pneumatic High Pressure Test Fixutre	BuWeps Dwg 2025362
Pressure Tank	BuWeps Dwg 2025374
Supports	BuWeps Dwg 2244646
Simulated Missile Skirt	SA-2440143
Structural Test Setup	SA-2440205
Corrosion Tank	SA-2440379

4.5 Production Tests

4.5.1 Responsibility - The contrctor or Government Inspector shall conduct production tests as specified in 4.3.1 to asure that the support ring assembly is in compliance with the requirements of 3.3.1 through 3.3.6.2.2.

4.5.2 Test conditions - Unless otherwise specified, the support ring assembly shall be subjected to production tests under the following conditions:

Temperature:	Room ambient, +65 to +85 degrees Fahrenheit
Altitude:	Normal ground atmospheric pressure
Vibraiton:	None
Humidity:	Relative humidity of 50 percent or less at +78 degrees Fahrenheit

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4.5.3 Test and Inspection Equipment and Facilities - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all production tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required production tests.

4.5.3.1 Calibration - Measuring equipment and instrumentation shall be calibrated as required by NAVORDINST 4355.1, BUWEPSINST 4355.5, and in accordance with the policies specified in the Standards Laboratory Information Manual issued by the Bureau of Naval Weapons Representative, Pomona, California.

4.5.4 Test Procedures

4.5.4.1 Operational Tests

4.5.4.1.1 Clamping Ring and Clamping Ring Actuator Test - To meet the requirements of 3.3.1.1 and 3.3.1.2, each clamping ring shall be tested using the simulated missile skirt (SA-2440143). The clamping ring shall be actuated with air or hydraulic pressure (as applicable) and shall clamp and unclamp the missile skirt as specified. With the system depressurized the clamping ring shall be proved capable of releasing the missile skirt by spring action.

4.5.4.1.2 Clamping Ring Position Indicator Test - To meet the requirements of 3.3.1.3, each clamping ring shall be connected to a test lamp circuit and operated. During operation the test lamp shall indicate the engaged and disengaged position of the ring.

4.5.4.1.3 Departure Switch Test - To meet the requirements of 3.3.1.4 each departure switch shall be connected to a test lamp circuit and the switch shall be operated by means of a suitable test fixture to indicate missile in place and missile away within the limits specified.

4.5.4.2 Dielectric Strength Test - To meet the requirements of 3.3.2, all electrical components shall be tested by applying a test voltage of 325-volts AC between joined terminals and a ground for 1 minute.

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4.5.4.3 Insulation Resistance Test - To meet the requirements of 3.3.3, all electrical components shall be tested by applying a test voltage of 500-volts DC between joined terminals and a ground.

4.5.4,4 Continuity Test - To meet the requirements specified in 3.3.4, all electrical components shall be tested by a point-to-point continuity check,

4.5.4.5 Working Pressure Tests

4.5.4.5.1 Type I - TO meet the requirements of 3.3.5.1, each support ring assembly, mounted in a suitable test fixture (Buweps Dwg 2025362 and 2244646), shall be tested bY applying a pneumatic proof pressure of 4,450 to 4,550 psig for 1 hour and shall not leak at a rate greater than 2 standard cubic inches per hour (1 cubic inch per cylinder).

4.5.4.5.2 Type II - To meet the requirements of 3.3.5.2, each support ring assembly shall be tested by applying a hydraulic proof pressure of 3,000 ± 150 psig for 1 hour and shall show no leakage. Wetting of the surface shall not constitute leakage.

4.5.4.6 Hydrostatic Pressure Tests

4,5.4.6.1 External - To meet the requirements of 3.3.6.1, each support ring assembly, mounted in a suitable test fixture (BuWeps Dwg 2025374), shall be tested by applying an external hydrostatic test pressure of not less than 580 psig for 30 minutes and shall show no leakage.

4.5.4.6.2 Internal

4.5.4.6.2.1 Type I - To meet the requirements of 3.3.5.2.1, all pneumatic- components of the support ring assembly, except the hoses, shall be tested by applying an internal hydrostatic test pressure of 6,700 to 6,800 psig for 2 minutes; the hoses shall be tested by applying an internal hydrostatic test pressure of 7,450 to 7,550 psig for 2 minutes. During testing, the components shall show no leakage or damage detrimental to continued operation.

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4.5.4.6.2.2 Type II - To meet the requirements Of 3.3.6.2.2, all hydraulic components of the support ring assembly shall be tested by applying an internal hydrostatic test pressure of 4,500 ± 100 psig for 10 minutes and shall show no leakage or damage detrimental to continued operation.

4.6 Preproduction Tests

4.6.1 Vibration Test, Operating Condition - To meet the requirements of 3.4.1, one support ring assembly shall be mounted in a suitable test fixture and shall be subjected to Type I vibrations specified in Standard MIL-STD-167 in the range of 5 to 25 cycles per second.

4.6.2 Vibration Test, Shipping Condition - One support ring assembly, mounted in a suitable test fixture, shall be tested to meet the requirements specified in 3.4.2 and shall show no damage.

4.6.3 Shock Test, Operating Condition - One support ring assembly, mounted in a suitable test fixture (SA-2440205), shall be subjected to the loads caused by a missile weight of 39,500 pounds accelerated as specified in 3.4.3, and thereafter shall be capable of meeting all performance requirements of this specification.

4.6.4 Inclination Test, Support Ring Assembly - To meet the requirements of 3.4.4, one support ring assembly, mounted in a suitable test fixture and having a simulated missile load of 39,500 pounds, shall be inclined to 60 degrees from the verticals under static conditions, and shall show no damage. The support ring, unloaded, shall then be subjected to inclinations up to 60 degrees from the vertical under dynamic conditions and shall show no damage.

4,6,5 Corrosive Atmosphere Test - To meet the requirements of 3.4.5, one support ring assembly shall be subjected to the salt spray test in accordance with Procedure I, Specification MIL-E-5272. Upon completion of the test, the support ring assembly shall be washed with fresh water and dried and shall show no corrosion detrimental to continued operation.

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4.6.6 Submersion Test - To meet the requirements of 3.4.6, one support ring assembly shall be subjected to submersion in tank (SA-2440379) of sea water for 90 consecutive days at temperatures between +28 and +100 degrees Fahrenheit. Upon completion of the test, the mechanism shall be washed with fresh water and dried, and thereafter shall be capable of meeting the performance requirements of this specification.

4.6.7 Heat/Pressure Cycle Test - One support ring assembly shall be installed in a Prototype launcher (Peashooter) or an acceptable equivalent test apparatus, shall be subjected to a heat/pressure test environment to meet the requirements specified in 3.4.7.2.

4.6.8 Operating Life Test

4.6.8.1 Clamping Ring, Type I - To meet the requirements of 3.3.7.1, one clamping ring actuated with air shall be operated for a total of 1,000 cycles of engagement and disengagement of the missile; the rate of cycling shall be such that the temperature rise of components will not exceed 15 degrees Fahrenheit. The clamping ring, upon completion of cycling, shall show no sign of wear detrimental to continued operation

4.6.8.2 Clamping Ring, Type II - To meet the requirements of 3.3.7.2, one clamping ring shall be actuated by hydraulic pressure and operated for a total of 2,000 cycles of engagement and disengagement of the missile; the rate of cycling shall be such that the temperature rise of components will not exceed 15 degrees Fahrerenheit. The clamping ring, upon completion of cycling shall show no signs of wear detrimental to continued operation.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Preservation and packaging of the support ring assembly shall be Level A or Level C as follows:

5.1.1 Level A - Level A shall be as specified by the procuring activity.

5.1.2 Level C - Cleaning, drying, preservation, and unit packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing - Packing of the support ring assembly shall be Level A or Level C as follows:

5.2.1 Level A - Level A shall be as specified by the procuring activity.

5.2.2 Level C - The support ring assembly, packaged as specified in 5.1.2, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classificatiion Rules and other common-carrier regulations applicable to the mode of transportation.

5.3 Marking - Unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

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NOTES

Intended Use

(a) The Type I support ring assembly is used in Launcher System Mark 17 Mod O with Launcher Tube Mark 4 Mod O and 1; in Launcher System Mark 17 Mod 1 with Launcher. Tube Mark 4 Mod 2 and 3.

(b) The Type II support ring assembly is used in Launcher System Mark 17 Mod 2 and Mark 21 Mod O and 2 with Launcher. Tube Mark 8 Mod O and 1.

6.2 Ordering Data - Procedurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Exceptions to the specification, applicable drawings, and other documents.

(c) Selection of applicable levels of preservation, packaging, and packing.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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